(TR-104) PYTHON WITH AI/ML

Training Day 29 Report

Date: 19 February 2026

Topic Covered: Resume Classifier AI Application, Word Embeddings & Semantic Resume Recommendation System

The twenty-ninth day of training focused on converting the NLP resume classifier into a deployable AI application and understanding modern NLP concepts including word embeddings, semantic similarity, and recommendation systems.

- Built a Resume Classifier AI application using Streamlit.
- Created an inference pipeline for resume category prediction.
- Saved trained Logistic Regression model and TF-IDF vectorizer using the joblib library.
- Implemented model loading and prediction workflow for real-time inference.
- Designed a Streamlit-based user interface for resume category prediction.
- Added validations for empty and short resume inputs.
- Tested the application using multiple real resume samples.
- Strengthened understanding of production-style AI application workflow.
- Learned the concept of word embeddings and semantic representation of words.
- Studied differences between TF-IDF and embeddings in NLP systems.
- Explored Word2Vec architecture including CBOW and Skip-Gram models.
- Used the Gensim library for Word2Vec training and similarity analysis.
- Learned about pre-trained embeddings such as GloVe and FastText.
- Understood sentence embeddings and semantic text representation.
- Implemented semantic similarity search using cosine similarity and cosine distance.
- Built a Resume Similarity and Recommendation Engine using embeddings.
- Generated embedding vectors for resumes and compared semantic similarity between resumes.

- Implemented top-k similarity search for retrieving most similar resumes.
- Generated category recommendations based on similarity matching results.
- Created reusable recommendation function for semantic resume matching and prediction.
- **GitHub Link:** https://github.com/JaspinderKaurWalia26/NLP/tree/main/Modern_NLP
- **Blog Link:** https://medium.com/@jaspinderkaurwalia855/turning-my-resume-classifier-into-a-real-ai-application-c266e7ecfe7c